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Application of Machine Learning to 3D Geomechanical Modeling for Estimating Resource Accessibility in the Permian Basin, USA

E. Liu, ExxonMobil Technology and Engineering Company, Houston, Texas, USA; N. M. Miller, S. Every, A. J. Dale, E. L. Ferreira, A. Bazzell, J. A. Bryant, and C. Lopez, ExxonMobil Upstream Company, Houston, Texas, USA; N. Kostov, X. Ma, and X.-H. Wu, ExxonMobil Technology and Engineering Company, Houston, Texas, USA

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Abstract

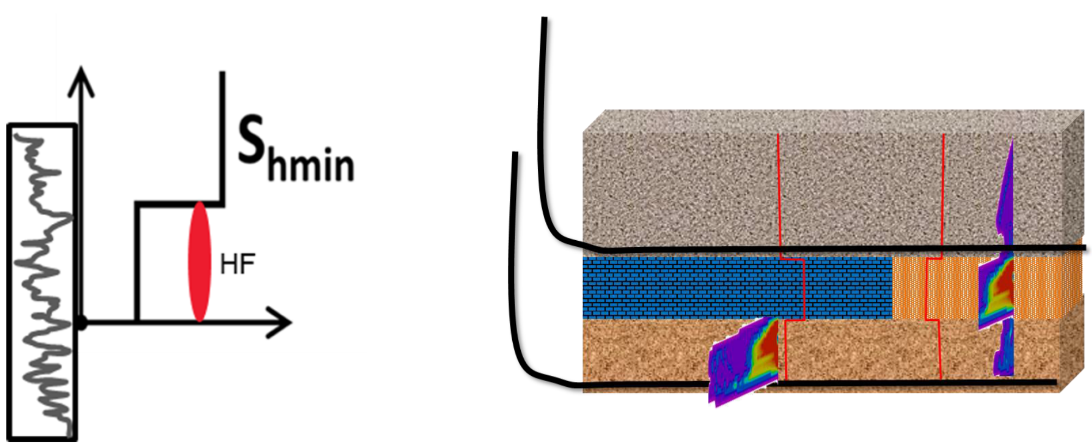
We employ a physics-based machine learning workflow that can be integrated with traditional geophysical analysis to improve estimates of stimulated rock volume in unconventional reservoirs, in addition to estimating resource density across flow units and benches, by significantly expanding the number of wells available for geomechanical analysis. Our workflow, tested in the prolific Permian Basin, USA, can accurately predict compressional-wave and shear-wave sonic data from existing log suites, expanding the number of wells available for geomechanical property calculation from hundreds to over tens of thousands; sonic logs can be predicted in wells that have only gamma ray, resistivity, and density logs available. Existing full-log-suite wells are used as a training dataset and for blind tests across all benches in the basin. The resulting predictions are used in calculations of various elastic properties and thus geomechanical properties, including minimum horizontal stress (S_{hmin}). The geomechanical properties are then used to populate a 3D model representative of subsurface variability across flow units and benches with an aim at refining cube performance (e.g. production rate) by bridging the gap between conventional estimated resource density and the stimulated rock volume (SRV). Seismic interpretation of key horizons can then be used to guide extraction of geomechanical properties from the model across the target intervals. The 3D geomechanical model can also be used for fracture modeling and for optimizing completion design. Our machine learning workflow is physics-based and scalable, which can be applied to thousands of wells. This enables us to rapidly understand past well performance, generate development scenarios, and guide future completions decisions. Thus far, the machine learning workflow has been tested in unconventional reservoirs, where S_{hmin} calculations are key to understanding the recoverable resource. The workflow itself has the potential to be extrapolated to other basins or scenarios to obtain a more robust understanding of subsurface properties impacting resource estimates and development plans on a large scale.

Introduction

Unconventional resource development requires accurate assessments of both resource density, which includes resource distribution and hydrocarbon pore volumes (HCPV), and resource accessibility, which includes well completion, ability of hydraulic fracturing and stimulated rock volumes (SRV). Both resource density and resource accessibility are critical for identifying potential development opportunities and optimizing resource extraction. Key parameters controlling resource density are thickness of hydrocarbon-bearing layers, porosity, and saturation. Resource accessibility is mainly controlled by geomechanics and state of in-situ stress. This study focuses on resource accessibility.

Understanding the in-situ state of stress is fundamental to all subsurface geomechanics and rock engineering endeavors, including drilling, hydraulic fracturing, enhanced oil recovery and induced seismicity (Johns et al., 2023; Amalokwu et al., 2023). Specifically, in unconventional resource development planning, it is necessary to know how the magnitude of horizontal principal stress varies from layer to layer. This affects drilling (wellbore stability), completions (hydraulic fracturing), saltwater disposal (mechanical seal capacity or caprock integrity), infill drilling and field development (parent-child well interactions, depletion planning) and abandonment (annular barriers).

Hydraulic fracture heights are directly controlled by S_{hmin}, as schematically illustrated in Figure 1a. It is important to emphasize that fracture heights are determined by the S_{hmin} difference between the landing zones and the overburden, not the absolute stress in the formation, although that is also important. For example, Figure 1b shows schematically the expected fracture height variations along a horizontal well when the overburden minimum stress varies laterally. When the overburden S_{hmin} is higher than in the landing zone (e.g., for a carbonate overburden), the hydraulic fracture is confined to the landing zone (ideal case); when the overburden S_{hmin} is lower than in the landing zone (e.g., in a weak shale overburden), the hydraulic fracture would extend beyond the landing zone into the overburden to create waste rock (which could result in parent-child interference if upper zones were developed later). Therefore, understanding lateral variation in lithology and minimum stress is important for optimizing completion. The magnitude and orientation of principal stress can be directly measured. The magnitude of the minimum principal horizontal stress (S_{hmin}) is obtained in the subsurface through identification of the closure pressure after shut-in of a diagnostic fracture injection test (DFIT), which involves the pumping of a relatively small volume of water into a short wellbore interval over a brief period (a minifrac test). The magnitude of the maximum principal horizontal stress (S_{hmax}) can be inferred from image log analysis of compressional failures (breakouts) and tensile failures (drilling-induced tensile fractures) of the borehole wall (Zoback et al., 2003). Neither measurement, however, is conducted at the frequency or scale needed for adequate characterization of horizontal stress magnitude variation with depth, or at the lateral frequency required for practical engineering applications across a region such as the Permian Basin.



(a) (b)

Figure 1—(a) Schematic illustration showing that minimum horizontal stress (S_{hmin}) controls heights of hydraulic fractures (HF). (b) Fracture dimension controlled by lateral lithology variation in both the landing zone and the overburden.

Multiple methods have been developed to predict stress away from points of direct measurement. Amalokwu et al. (2023) describe a new approach to calculate the horizontal principal stress state of the subsurface, with calibration constants defined by well-log analysis and seismic-scale facies. This allows the estimation of depth-varying horizontal stress profiles away from wells to populate the distribution of subsurface principal stresses in a 3D mechanical earth model (MEM). There also are several model-driven approaches that have been proposed to predict variation of subsurface stresses with rock type and depth. For example, under the assumptions of poroelastic coupling and anisotropic symmetry (typically vertical transverse isotropy, or VTI), horizontal stress magnitude variation with depth can be quantified from log-derived density, acoustic velocities, mineral composition, and pore-pressure properties, plus additional "tectonic strain" calibration to subsurface measurements such as DFIT or instantaneous shut-in pressure (ISIP). Estimating stress in a 3D model can be difficult away from well control. Johns et al. (2023) present a novel workflow for simplifying theoretical stress calculations and calibrating a seismic-driven model to log-based 1D S_{hmin} profiles. The workflow employs these calibration constants to propagate stress magnitudes throughout the 3D model based on 3D properties such as overburden stress and reservoir or pore pressure, as well elastic properties derived from seismic inversion and rock facies. The 3D MEM produces S_{hmin} profiles, maps, and cross-sections that can be used for assessing stress across a development region such as the Permian Basin.

This paper presents a new workflow to predict elastic properties, including S_{hmin}, that enables the construction of detailed 3D geomechanical models. After establishing the link between elastic and geomechanical properties, we describe the workflow, including discussion of examples and validation, and then we present applications.

The Link Between Elastic Properties and Geomechanical Properties

For an anisotropic poroelastic formation in which total principal horizontal stresses arise from gravitational loading (overburden and pore pressure) in the vertical (z) direction, with additional stress components from tectonic strain in the horizontal (x-y) plane, the minimum and maximum subsurface stresses (S_{hmin}, S_{hmax}) can be derived according to Hooke's law:

$$S_{hmin} = \left[\left(E_H / E_V\right)\nu_V / \left(1-\nu_H\right)\left(S_{OVB} - \alpha_V P_{RES}\right) + \alpha_H P_{RES}\right] + \left[E_H / \left(1-\nu_H^2\right)\varepsilon_{HMIN} + E_H \nu_H / \left(1-\nu_H^2\right)\varepsilon_{HMAX}\right] \quad (1)$$

$$S_{hmax} = \left[\left(E_H / E_V\right)\nu_V / \left(1-\nu_H\right)\left(S_{OVB} - \alpha_V P_{RES}\right) + \alpha_H P_{RES}\right] + \left[E_H / \left(1-\nu_H^2\right)\varepsilon_{HMAX} + E_H \nu_H / \left(1-\nu_H^2\right)\varepsilon_{HMN}\right] \quad (2)$$

where S_{OvB} = overburden stress, P_{RES} = reservoir or pore pressure, E = Young's modulus, V = Poisson's ratio, a = Biot's coefficient (or effective stress coefficient), ε_{HMIN} = minimum principal horizontal strain, and ε_{HMAx} = maximum principal horizontal strain; subscripts v and H refer to vertical and horizontal, respectively. The second term in both equations (1) and (2) is usually called the tectonic strain components. In the case of isotropy, i.e., no directional dependence of Young's modulus, Poisson's ratio, or Biot's coefficients, S_{hmin} and S_{hmax} depend only on Poisson's ratio. Note that the limitations and validity of the assumption used in deriving equation (1) are discussed by Iverson (1995) and Katahara (2009). We assume it is applicable to the Permian Basin (Johns et al., 2023; Amalokwu et al., 2023) as well as other unconventional resource plays (e.g., Zhang, et al., 2023). A detailed discussion of Biot's coefficients, tectonic strain and anisotropic calibration in this equation is beyond the scope of this paper; see Johns et al. (2023) and Amalokwu et al. (2023).

For a typical 1D workflow based on wireline logs, acoustic and density logs can be used to determine isotropic dynamic elastic properties, which are relatively high-frequency and low-strain: Young's modulus (EDYN) and Poisson's ratio (VDYN). These dynamic properties are then calibrated to direct core-plug measurements of anisotropic static elastic properties, which are relatively low-frequency and low-strain: Young's modulus (E_v, E_H) and Poisson's ratio (V_v, V_H). Mineralogical composition is typically determined either from elemental spectroscopy logs or through multimineral inversion of standard "quad combo" logs (gamma ray, resistivity, neutron-density, photoelectric factor, compressional-wave and shear-wave slowness). This composition is then combined with the static elastic properties to calculate associated lithology-dependent anisotropic poroelastic properties: Biot's coefficient (α_v, α_H). For discussion on reservoir and pore pressure, see Sayers and den Boer (2019) and vernik and Newton (2022).

Static elastic and poroelastic property logs can then be combined with pore pressure (from direct downhole measurement or various log- and drilling-based prediction techniques) and overburden stress magnitudes (most quantified through density log integration) to define the first term in equations (1) and (2). This gives an initial, uncalibrated horizontal stress versus depth profile. The predicted profile can then be adjusted into congruency with direct measurements of subsurface S_{hmin} magnitude (typically from DFIT) by optimizing the £HMIN and ε_{HMAx} tectonic strain constants in the second term of equation (1), generating a final, DFIT-calibrated S_{hmin} profile. Applying the same calibration to equation (2) generates an equivalent S_{hmax} profile. Alternatively, direct quantification of S_{hmax} magnitude could be made from borehole breakout and/or drilling-induced tensile fracture analyses incorporating image logs and formation strength (Zoback et al., 2003).

The dynamic Young's modulus and Poisson's ratio are fundamentally linked to the compressional-wave velocity and shear-wave velocity, which are the inverse of sonic logs: compressional-wave slowness (DTC) and shear-wave slowness (DTS). Figure 2 summarizes the key equations that link the seismic or elastic properties in terms of compressional-wave and shear-wave velocities (v_p and v_s, respectively) to geomechanical properties in terms of Young's modulus (E) and Poisson's ratio (v). Both sets of parameters are fundamentally linked to bulk modulus (K) and shear modulus or rigidity (^). It is important to note that properties derived from seismic or well log data are so-called dynamic properties (measured at very small strain and small stress) and need to be converted to static properties (measured at large strain and large stress). The transformation between dynamic and static properties is usually derived from laboratory data. The minimum horizontal stress is computed from equation (1), based on a 1D mechanical earth model assuming uniaxial stress.

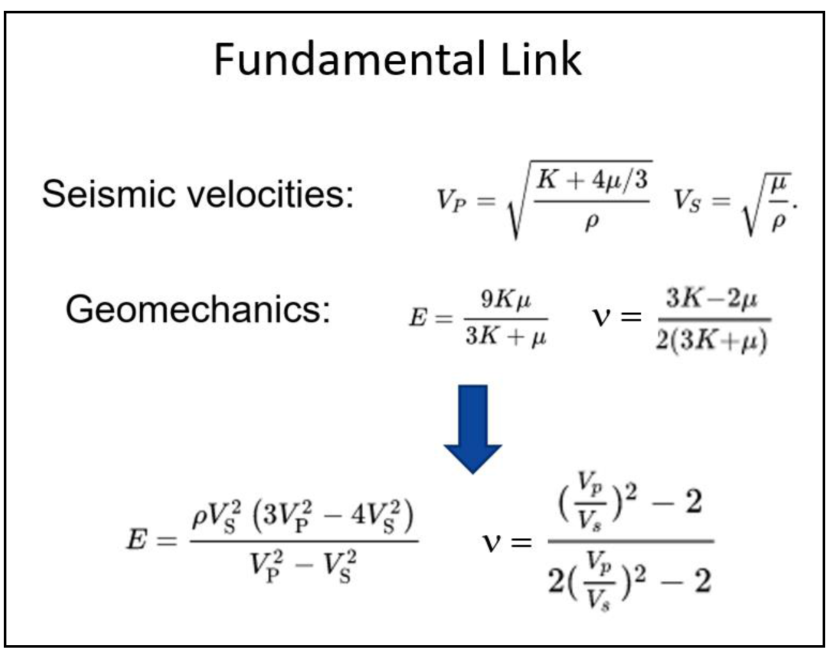


Figure 2—Fundamental links between seismic- and sonic-derived compressional-wave (V_p) and shear-wave (V_s) velocities and dynamic geomechanical properties in terms of Young's modulus (E) and Poisson's ratio (v).

About 2% of the wells in the Permian Basin have the compressional-wave and shear-wave sonic data required to generate S_{hmin} curves. This dataset is at least one to two orders of magnitude smaller as compared to other subsurface attributes (e.g., hydrocarbon pore volume, or HCPV), which limits regional mapping of geomechanical variation. The traditional way of predicting sonic properties is to use a rock physics model (such as the method for organic-rich shale described by Zhu et al., 2011), which requires inputs such as minerology (calcite, dolomite, quartz, clay, total organic carbon) and associated elastic properties (bulk and shear moduli as well as bulk density), porosity and saturations. Such rock physics modeling (e.g., Xu and Payne, 2009; Zhu et al., 2011; Sayers, 2019; Vernik, and Milovac, 2011) is usually a manual process and very time-consuming, although it can be optimized and automated. To map geomechanical and stress variation regionally, this kind of rock physics modeling is not generally not practical because it would require predicting compressional-wave and shear-wave sonic properties for thousands of wells. Recently, there have been some attempts to apply machine learning and neural networks to predict shear-wave velocities and facies (e.g., Vento et al., 2023; Yamatani and Desaki, 2023), and conduct petrophysical modeling in general (e.g., Bhattacharya, 2022; Wang, 2024),. In this paper, we propose a machine learning approach to sonic prediction so that every triple-combo well can potentially be utilized to constrain mechanical properties and stress magnitudes, enabling regional-scale, well-calibrated mechanical earth models.

Machine Learning Workflow for Sonic Prediction

This paper aims to provide a mechanism for sonic-log prediction that combines physics-based constraints and data-driven features. This workflow has the capability to predict hundreds and even thousands of sonic logs in a matter of minutes, which will accelerate turnaround in the process of unconventional asset development and enable real-time well-to-seismic ties and borehole-stability estimation for conventional assets. The physics-based constraints ensure that model output is plausible and aid in interpreting the results. The workflow includes raw data ingestion, data quality control with physics constraints, a hierarchical model prediction pipeline, hierarchical model training and retraining, quality control of prediction results, uncertainty quantification, confidence map generation, and output file generation. The workflow is very robust in terms of handling missing values, scaling accurately y to the number of input logs, and generating insights about the influence of data-driven features on model predictions, which aids interpretation of the results. The workflow uses the SPARK engine (Zaharia, 2016) for large-scale data manipulation, model building and inferencing.

The machine-assisted sonic prediction workflow includes these steps:

1. *Raw data preparation*
   a. Ingest raw well log data in LAS format and maintain the pipeline of data extraction, transformation, loading (ETL).
   b. Automatically select the best available source of input logs. (It is strongly recommended to work together with formation evaluation experts.)
2. *Data quality control with physics constraints*
   a. Incorporate physical constraints to remove erroneous data and outliers from the model training dataset. For example, check the V_p/V_s ratio to make sure the measured V_p and V_s are within physical ranges.
   b. Remove outliers in the training data to reduce bias (data-driven approach to quality control). For example, de-spike the caliper data.
3. *Hierarchical machine learning model training*
   Use the chain regression model with XGBoost (Chen and Guestrin, 2016) as the back-end modeling engine. (This selection was made after testing various machine learning algorithms.) The chain

regression model can provide robust predictions with wells that lack sonic data. The well data inputs are triple-combo logs: Gamma ray (GR), deep resistivity (RESD), bulk density (RHOB), neutral porosity (NPHI) and/or photo-electric (PE). These data are used to predict compressional-wave sonic (DTC) logs, which are then used with the other logs to predict shear sonic (DTS) logs, based on the typically strong correlation between DTC and DTS. The measured depth (DEPTH) curve is also included to capture the compaction trend, although this is not necessary.

4. *Hyper-parameter optimization for machine learning model*

   a. Perform a search (grid or random) over the hyper-parameter values in order to identify optimal values for chain regression with XGBoost model as the back-end using cross-validation. Important hyper-parameter values for tree-based models are the number of estimators and the minimum number of samples to split.
   b. A rock physics model (Xu and Payne, 2008) and/or physical bounds (e.g., V_p/V_s (or DTS/DTC) > 1.414, or, equivalently, $0 < v < 0.5$) can be used to constrain the predicted sonic values if multi-mineral logs for the corresponding wells are available.

5. *Data separation for training and testing*

   Select wells by region to train and validate the model with data that are adequately representative.

6. *Compressional-wave and shear-wave sonic log prediction from the trained model*

   Accurate prediction requires the training data to be representative of subsurface geology, lithology, etc., of the study area.

7. *Quality control of post-prediction output*

   The above sequence is implemented as a batch workflow and can be used to predict compressional-wave and shear-wave sonic for many thousands of wells. It is also a parallelization prediction workflow to handle high-traffic data loads. The workflow is shown graphically in Figure 3.

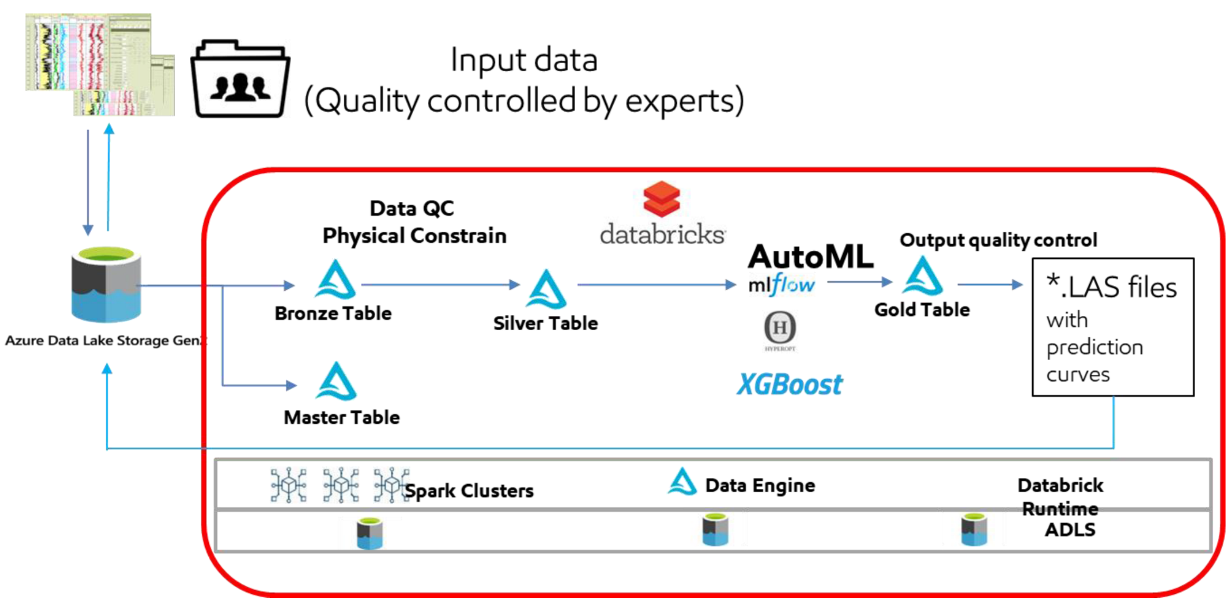


Figure 3—Machine-assisted sonic prediction workflow. Input data are GR, RHOB, RESD, NPHI and/or PE. Output logs are predicted compressional-wave and shear-wave sonic logs (DTC and DTS, respectively). Input data need careful quality control (QC) by formation evaluation experts.

Examples and Validation

We show three examples of blind model tests from the Permian Basin to validate the workflow. The wells in these examples are not used in the training dataset. In the first example, shown in Figure 4, measured DTC data are used in addition to GR, RHOB, RESD, and NPHI data as inputs to predict shear sonic logs. The normalized absolute root mean square (NRMS) error between the measured and predicted V_s is consistently less than 5%, with exceptions only at the very few depths the measured log has spikes not present in the predicted log, which is smooth and consistent with other input logs. The average NRMS error is less than 2%.

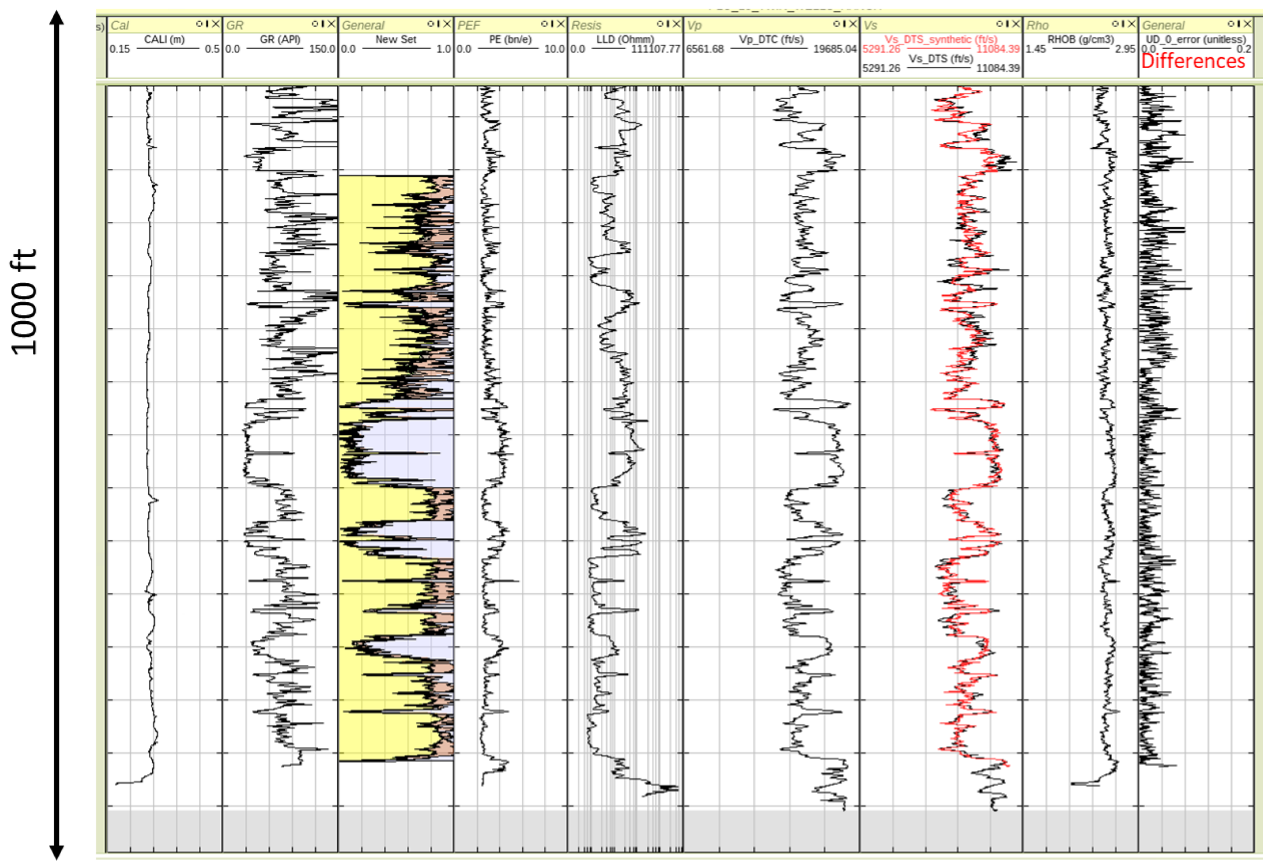


Figure 4—An example with predicted shear sonic (DTS) overlain in red on measured data (black) in the seventh column. The last column shows the normalized absolute root mean square (NRMS) difference between the measured and predicted Vs in fractional form. Note that the difference is consistently less than 5% with an average of about 2%.

The second example, shown in Figure 5, includes predicted logs for both V_p and V_s. Measured V_s is available only for the deep part of the well; the predicted V_s successfully fills that deeper section and matches the recorded sonic data there well.

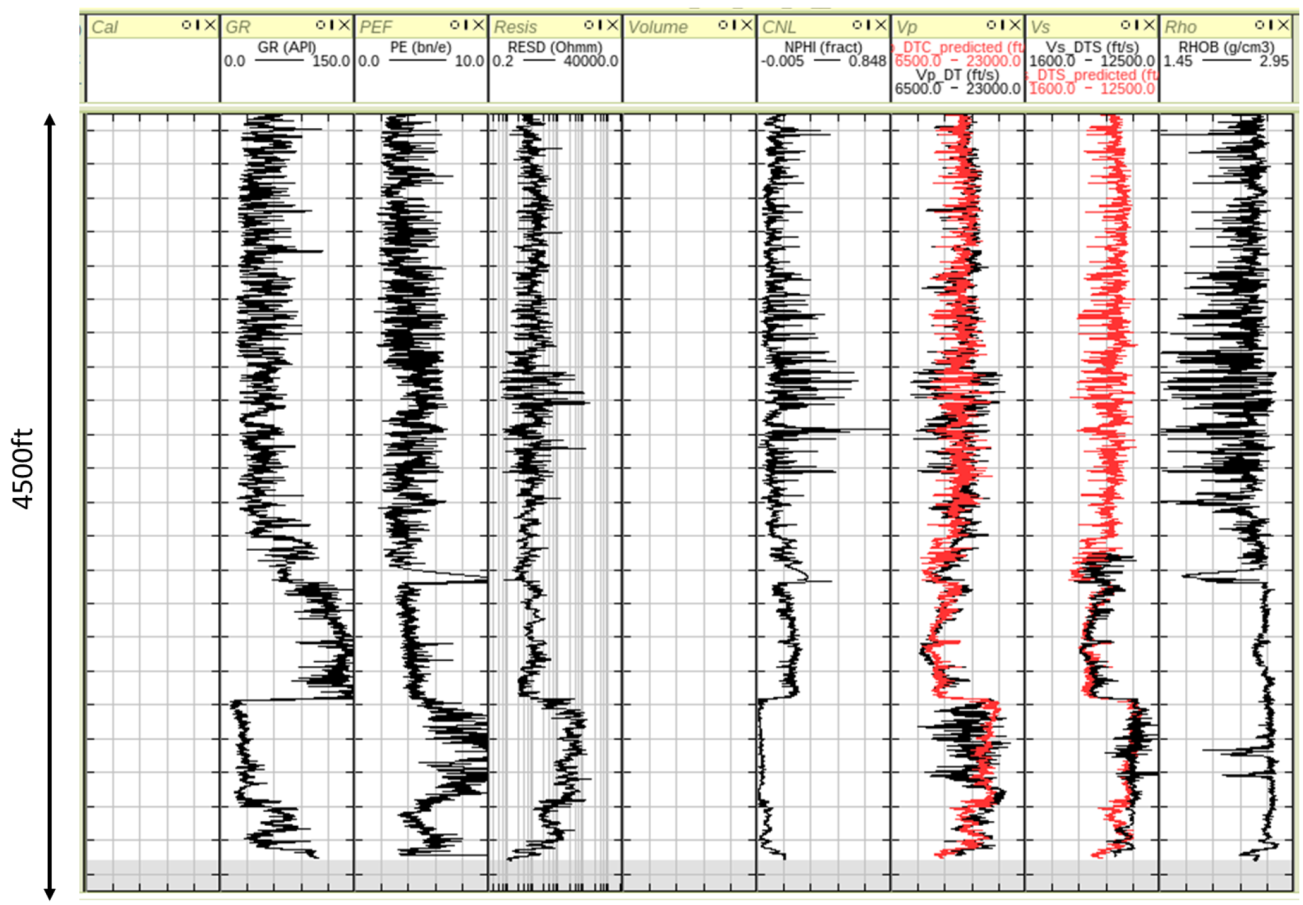


Figure 5—An example with predicted compressional-wave sonic (DTC) and shear sonic (DTS) overlain in red on measured data (black) in the seventh and eighth columns, respectively. Note that measured shear sonic data are available only for the lower part of the well, and the predicted logs (red) fill the missing shear sonic data.

The third example, shown in Figure 6, demonstrates the importance of quality control in measured logs, particularly Vs, which can be checked by comparing the V_p/V_s ratio or Poisson's ratio with the physical band and with adjacent wells. The Poisson's ratio derived from measured logs (shown in the last column) reaches an average value of 0.4, which is too high for typical rocks in the Permian Basin. (A high Poisson's ratio of 0.4 is typical of materials such as mashed potatoes or beach sand.) The Poisson's ratio from the predicted sonic data, in contrast, is within the appropriate physical range and consistent with the nearby wells.

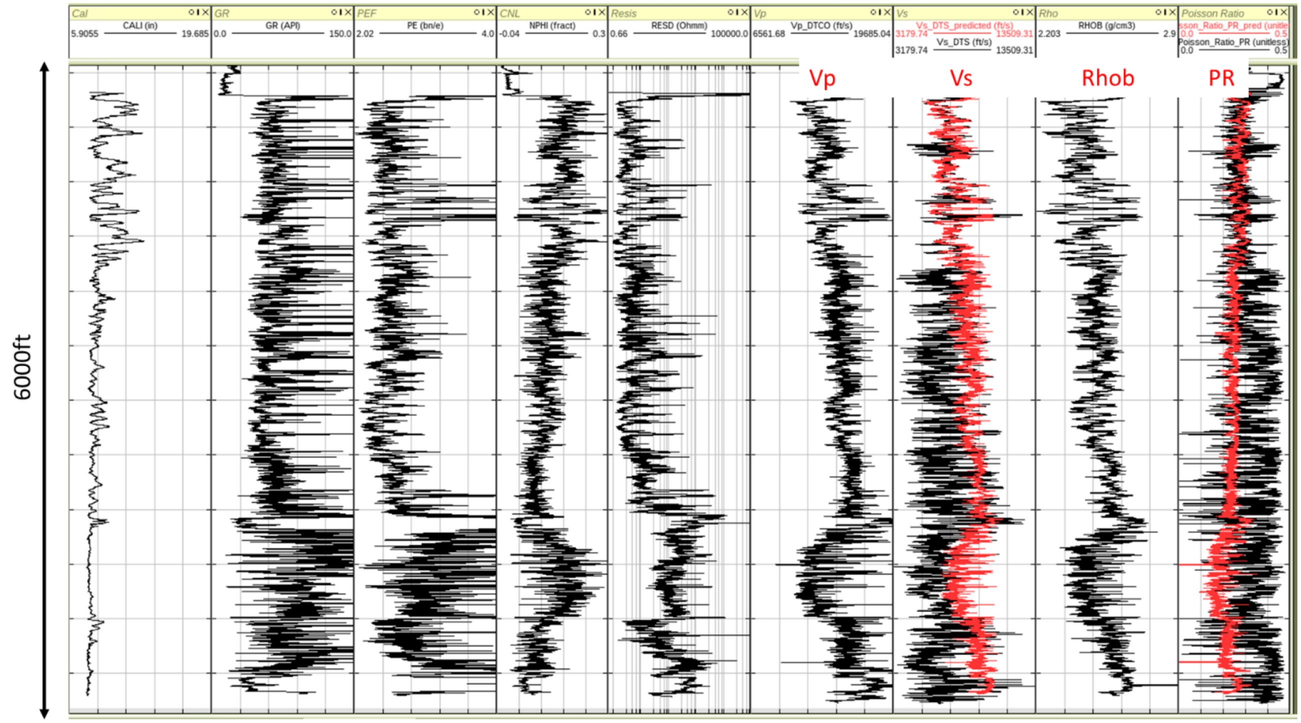


Figure 6—An example with predicted shear sonic DTS (red) overlain on measured data (black) in the seventh column. There are large differences that are due to errors in measured logs, as indicated by anomalously high Poisson's ratio (PR) derived from those logs, shown in black the last column.

3D Mechanical Earth Model Workflow

Once the compressional and shear sonic logs are predicted for thousands of wells, they can be used to build 3D geomechanical models using the method as described by Johns et al. (2023). Our model building offers the opportunity to use not just tens but many thousands of wells. Seismic data can potentially be incorporated if available, but may provide minimal uplift in areas with abundant wells.

Figure 7 summarizes our 3D mechanical earth model workflow. We start by constructing a model framework for the area of interest and then create the model grids. The model grids can be coarse or fine, depending on the project objective. For example, a regional study would call for a coarse grid, e.g., 500 ft × 500 ft × 25 ft, whereas a local development area would call for a fine grid, e.g., 50 ft × 50 ft × 10 ft. The next step is to extrapolate the properties from wells (using any of the widely available methods). Finally, we convert the model to a seismic grid, so that standard seismic interpretation tools can be applied to extract surface and interval properties. Quality control is important at every step, as shown in the flow chart in Figure 7. Note that the accuracy of the 3D properties will depend on well density and well distribution, and will likely be low where wells are sparse.

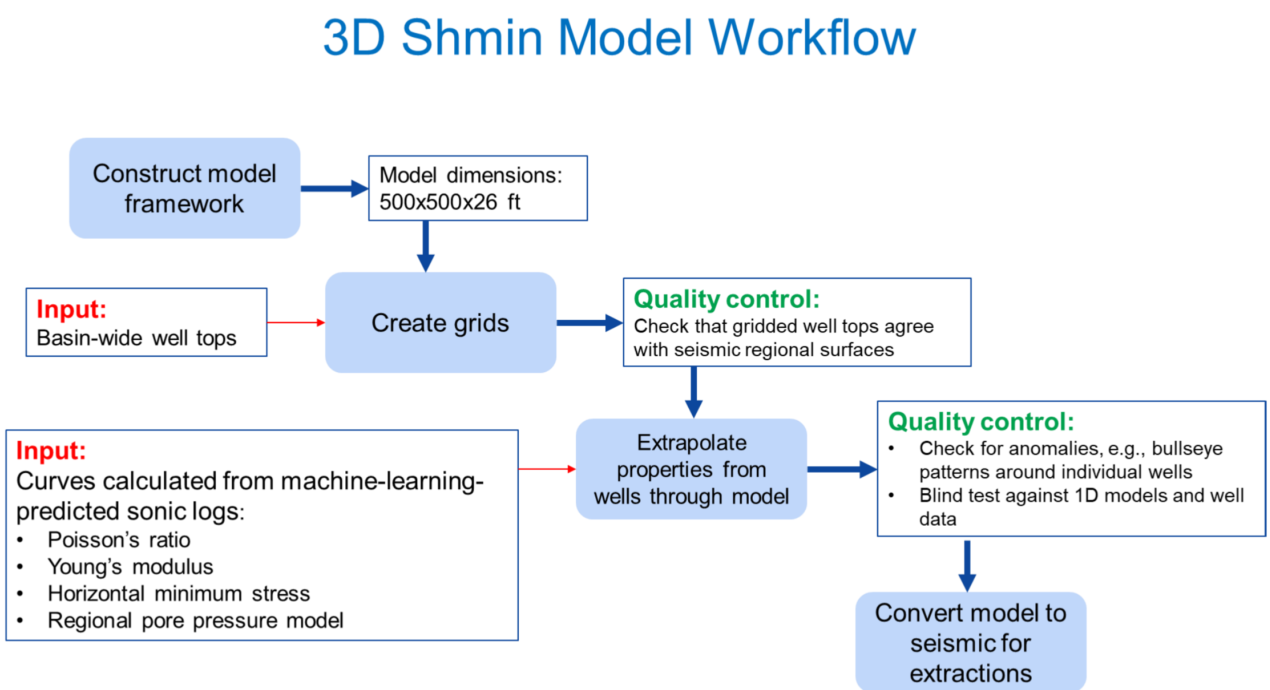
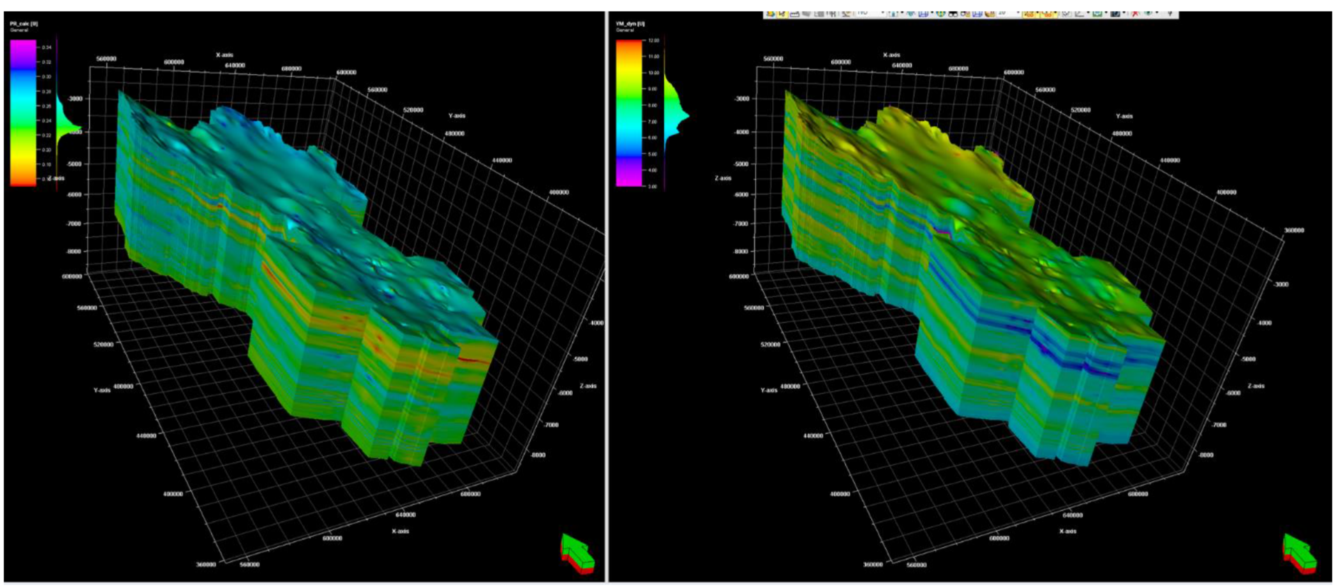


Figure 7—Proposed workflow to build well-based 3D mechanical earth models using predicted sonic data.

Figure 8 shows an example of a well-based 3D geomechanical model in the Permian Basin. The 3D predictions of Poisson's ratio (left) and Young's modulus (right) can be used for development planning.



(a) (b)

Figure 8—Well-based 3D geomechanical model from the Permian Basin: (a) Poisson's ratio and (b) Young's modulus

Applications

Once a 3D model is built for mechanical properties (Young's modulus and Poisson's ratio) and for minimum horizontal stress (S_{hmin}), it can be applied to a variety of technical needs. Two potential applications are reviewed here: analogue well selection and completion design and optimization.

Analogue well selection

For unconventional resource development, it is often necessary to find an analogue well for fracture modeling and optimizing completion design. How best to choose the analogue well remains an open question, but it is commonly recommended to use the nearest well available. Figure 9 shows why this approach could result in an inappropriate choice, however. In this example, well A is nearest the development area, but well C is the better analogue in terms of predicted fracture behavior. We recommend that S_{hmin} differences between overburden and landing zones, not just well proximity, should guide analogue well selection.

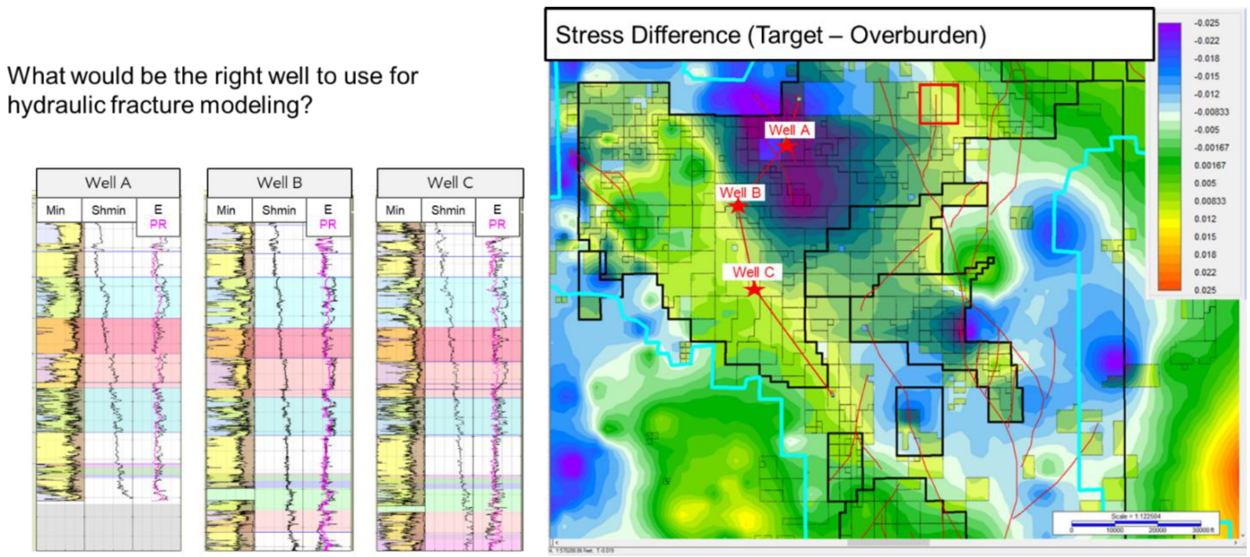


Figure 9—Example of analogue well selection. Right: The red box shows the development area, overlain on a shaded map of stress (S_{hmin}) difference between the target landing zone and overburden. Cool colors show where S_{hmin} is larger in the overburden than in the target landing zone, implying that hydraulic fractures would be confined to the landing zone (ideal case). Warm colors show where S_{hmin} is smaller in the overburden than in the target landing one, implying that the hydraulic fractures would extend beyond the target interval; this could result in parent-child interference during potential future development of higher sections. Left: Key logs for wells A, B, and C.

Completion design and optimization

3D geomechanical models can also help optimize completion design along the long (5,000-20,000 ft) horizontal wells that are typical of unconventional developments. These wells are commonly stimulated by fracking in sections (stages) that measure 200-500 ft from toe to heel. Fracture networks can be highly complex and variable along lateral length of these wells. 3D mechanical models can help address this complexity by differentiating optimal and sub-optimal stages on the basis of graded completion data.

The quality of completions depends on several elastic properties, which vary laterally within individual benches. Table 1 summarizes key completion parameters and mechanical properties that have geological and lithological relevance. Note that these properties include not only Young's modulus and Poisson's ratio, but also Rickman's brittleness (B), which is defined as a combination of scaled Young's modulus (E) and Poisson's ratio (ν):

$$E_{brit} = \left(E_{dyn} - 1\right)/7, \tag{3}$$

$$\nu_{brit} = \left(0.4 - \nu_{dyn}\right)/0.25, \tag{4}$$

$$B = \left(E_{brit} + \nu_{brit}\right)/2. \tag{5}$$

All quantities in the above equations are dynamic properties and a similar definition can be applied to static brittleness. Given potential sensitivity to vertical and lateral variation in the properties, their vertical and lateral derivatives are also included in Table 1.

Table 1—(a) Key completion parameters, excluding engineering parameters such as borehole penetration azimuth, lateral length, number of stages, fracture spacing, etc. (b) Key mechanical properties that have geological and lithological relevance. Note that dZ is the derivative with respect to vertical depth (Z), and dL is the derivative with respect to lateral (horizontal) length (L).

(a) Key completion parameters	(b) Key mechanical properties
Proppant variation from design Clean volume variation from design Treatment pressure/injection rate ISIP variation from S_{hmin}	Young's modulus (E) Poisson's ratio (v) Brittleness (B, scaled E and v) S_{hmin} gradient (dS_{hmin}/dZ) dE/dZ (vertical gradient in E) dE/dL (lateral variation in E) dB/dZ (vertical gradient in B) dB/dL (lateral variation in B)

Figure 10 shows the lateral distribution of elastic parameters along wells within the same unit and bench. Each property has a wide distribution over short distances, supporting the hypothesis that elastic properties impact completions. Geomechanical properties for Formation A (upper bench) and Formation B (lower bench) are different, indicating different expectations for behavior in completions.

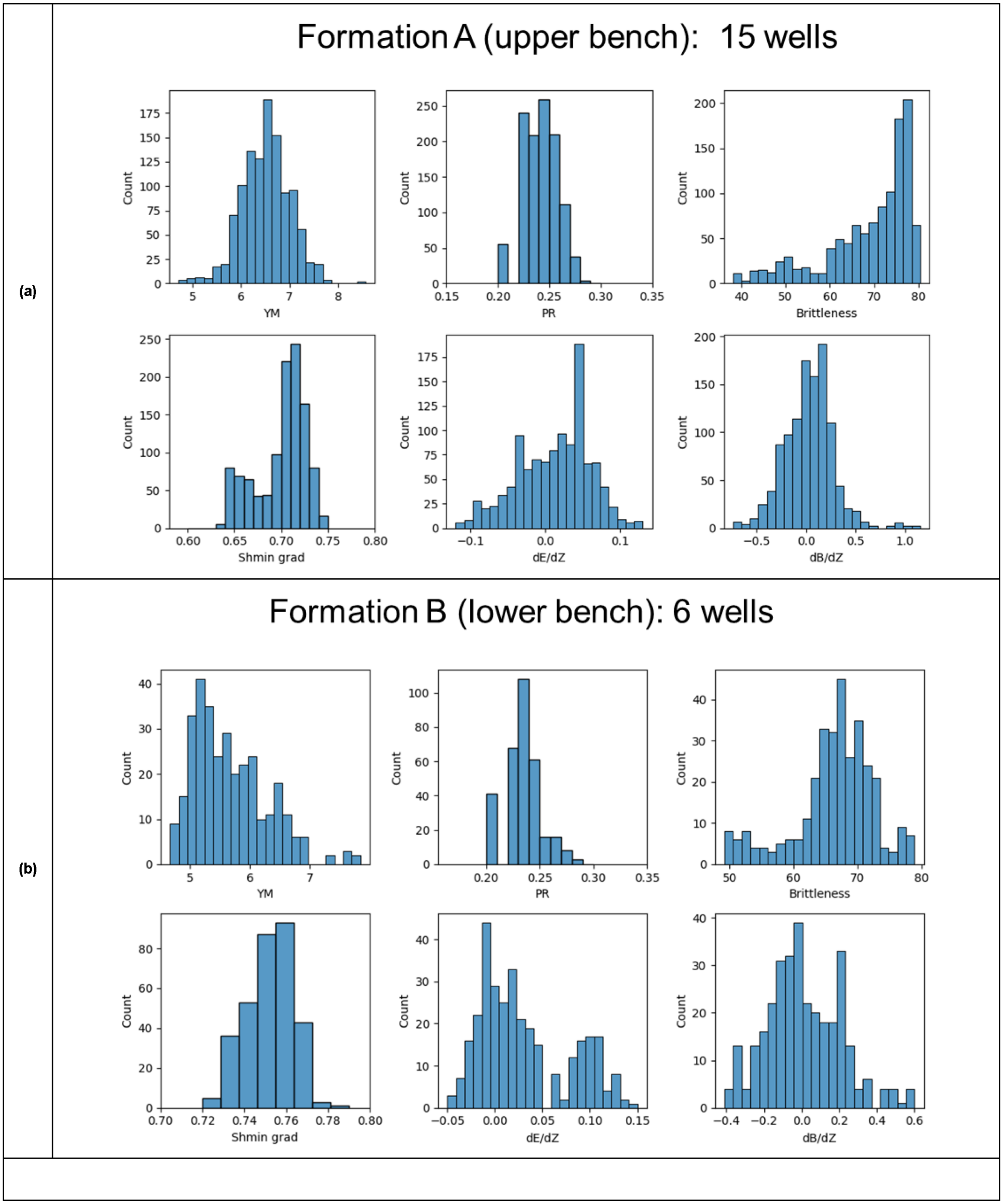


Figure 10—Lateral distribution of elastic parameters along wells within the same unit and bench: (a) Formation A (upper bench); (b) Formation B (lower bench). YM: Young's modulus. PR: Poisson's ratio. Shmin gradient, gradient in minimum horizontal stress.

An example of these differences in behavior is shown in Figure 11. It depicts a hydraulic fracturing experiment at two different stages, one in which fracturing is inconsistent, and fluid pressure cannot be maintained (unstable behavior), and one in which fluid is pushed into the formation at fracturing pressures and not lost to high-permeability channels (stable behavior).

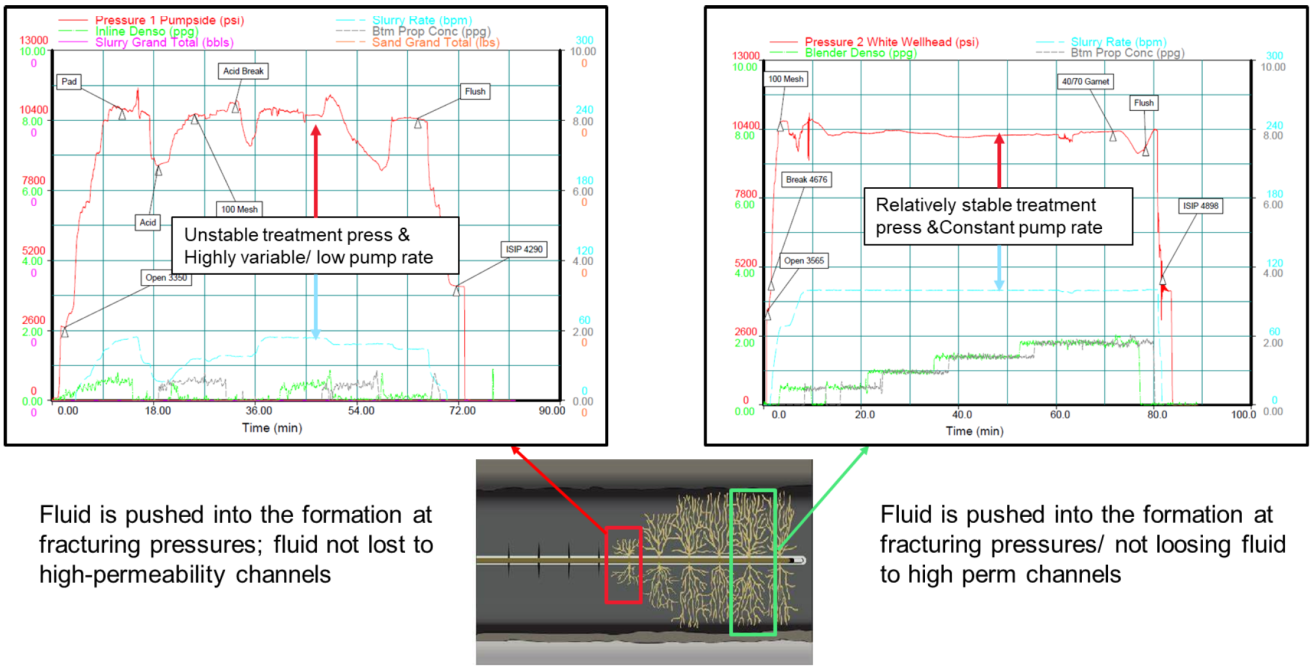


Figure 11—An example of a hydraulic fracturing experiment at two different stages. Left: Unstable treatment pressure and highly variable, intermittently low pump rate. Right: Relatively stable treatment pressure and constant pump rate.

With this variation in behavior in mind, we grade completion stages as follows:

- Good stages follow through on design parameters - little to no deviation
- Fair stages show small anomalies in execution pressures, pumped volumes, and pumping rates
- Poor stages show significant variability in pumping rates or well pressures as well as pumped volumes

Each grade corresponds to a set of property ranges, which were constrained using a random forest classification. This classification was applied to predict completion stage scores for Formations A and B from the example above. Figure 12 shows the rankings of mechanical properties - features - based on relative influence on the results. Brittleness and Young's modulus are generally the largest contributors. Preliminary results show that model predictions of good stages tend to be accurate, but model predictions of poor stages are less accurate.

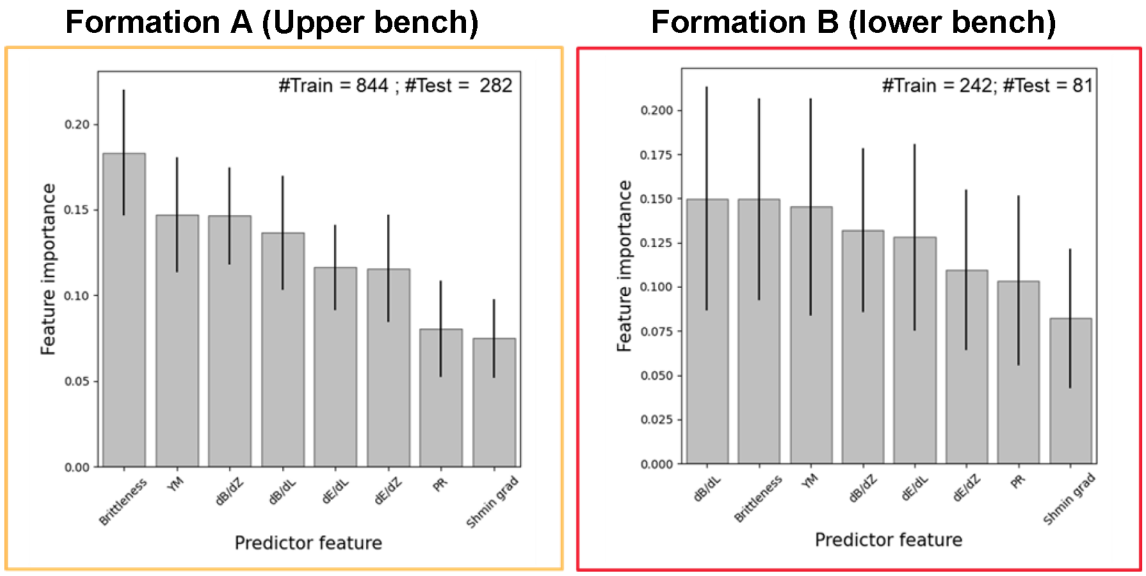


Figure 12—Feature importance from the random forest classification, which provides an ensemble of decision trees (averaging over models to reduce bias) to predict completion stage scores. YM: Young's modulus. PR: Poisson's ratio. S_{hmin} gradient: gradient in minimum horizontal stress.

Conclusions

We have applied a physics-based machine learning workflow to predict elastic properties for thousands of wells, enabling the construction of detailed 3D geomechanical models. We integrated the 3D geomechanical models with traditional well-based and seismic-derived estimates of hydrocarbon pore volume, allowing us to see expected fracture performance for zones of net pay. This integration adds insight into resource accessibility to conventional resource calculations. Addressing both resource density and resource accessibility supports the generation of development scenarios, enhances understanding of historical well performance, and informs future completion strategies. The workflow has been successfully tested in unconventional reservoirs, where geomechanical properties, particularly minimum horizontal stress, are essential for evaluating resource potential. We have shown two applications, one for analogue well selection and completion design/optimization. The methodology is broad enough to be adapted for other applications, such as predicting facies or classifying environments of deposition. This flexibility makes the workflow a powerful tool for enhancing resource estimates and guiding development planning across diverse geological settings.

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